THE CURE CENTER
FOR ULTRASOUND RESEARCH & EDUCATION

OBSTETRICS SYLLABUS

COURSE CODE: DMS 305
OBSTETRICS

Instructor:
Telephone:
E-Mail:
Office Hours:
Mon, Wed, Fri: 12pm to 1pm & 5pm to 6pm
Tues: 9am – 5pm
Thurs: 11am – 5pm

Start Date:

Completion Date:

Days and Times:
Monday, Wednesday and Friday

Course Hours:
Morning Session:
8:00 am -10:00am &
10:10am – 12:00 pm
Afternoon Sessions:
1:00pm – 3:00pm &
3:10 pm – 5:00 pm

Class Location:
Classroom #:
Lab#:

Prerequisites:
BIO 100 Introduction to A&P
PAT 100 Patient Care
LAW 100 Medical Law & Ethics

Course Contact Hours: 72
Lecture: 32 hours
Laboratory: 40
Course Delivery: Residential
Pre-requisite: None

Americans with Disabilities Act Guidelines:
Requests for accommodations should be submitted to the campus president. Faculty are not authorized to grant accommodations.

COURSE DESCRIPTION

This is a course directed towards developing the principles techniques, knowledge, and skills of each student necessary to function as a qualified diagnostic medical sonographer in clinical and diagnostic practice of obstetrics sonography.

In this course will discuss various aspect of obstetrics sonography. The course content includes detailed study of the anatomy and physiology of the fetus visible by ultrasound imaging, as well as pathologic states met during transabdominal, transvaginal, and trans-labial ultrasound evaluation of the female pelvic.

Student will learn and comprehend the embryologic development, development of the placenta, the different trimesters that occur in a pregnancy and what you see in each trimester.

Student will learn to accurately survey and perform the obstetrics sonograms and will collect a comprehensive and thorough knowledge of the anatomy physiology pathology and the sonographic appearance of the female reproductive system.

Student will gain proficiency in the diagnostic ultrasound evaluation for each trimester. They will be able to recognize what is normal anatomy for each trimester and recognize pathology.

Emphasis will be placed on the academic and laboratory applications within the highly specialized area of obstetrics sonography including basic patient care principles and professionalism.

Instructional Materials

1. Textbook
Title: Textbook of Diagnostic Sonography, 8th Edition (2 Volume Set)
Author(s): Sandra L. Hagen-Ansert, MS, RDMS, RDCS, FASE, FSDMS
Publisher: Mosby; 8th edition (May 3, 2017)
ISBN: 978-0323353755

2. Workbook
Title: Workbook for Diagnostic Sonography, 8th Edition
Author(s): Sandra L. Hagen-Ansert, MS, RDMS, RDCS, FASE, FSDMS
Publisher: Mosby; 8th edition (May 3, 2017)
ISBN: 978-0323441834

3. Reference Book
Title: Pocket Protocols for Sonography Scanning, 4th Edition
Author(s): Betty Bates-Tempkin, BA, RT®, RDMS
Publisher: Saunders; 4th edition (December 27, 2014)
ISBN: 978-1455773220

Equipment/Technology

- LCD Projector
- Laptop Computer
- Mindray Ultrasound Machine

Material

- Electronic and Printed Worksheets
- Textbooks
- Visual Aids: DVD Videos, Case Study Reviews, Demonstrations and Hands on Clinical Laboratory Practice.

During clinical laboratory sessions various techniques such as role playing and scanning exercises will be used.

OBSTETRICS SYLLABUS

Grade Point Value

Grade	Percentile Range	Points
A+	97-100	4.00
A	93-96.99	3.85
A-	90. -92.99	3.67
B+	87-89.99	3.33
B	83-86.99	3.00
B-	80-82.99	2.67
C+	77-79.99	2.33
C	73-76.99	2.00
C-	70-72.99	1.67
D+	67-69.99	1.33
D	63-66.99	1.00
D-	60-62.99	.67
F	Below 60-00	0

Methods of Evaluation

Exams / Quizzes/ Midterm/ Final	40%
Projects/ Attendance Participation/ Homework Assignments	20%
Scan-out	40%
	100%

Teaching Strategies

A variety of teaching strategies may be utilized in this course, including but not limited to lecture, discussion, written classroom exercises, visual aids, videos, case study reviews, group and/or individual projects, quizzes, exams, reading assignments, writing assignments, interactive exercises demonstrations and hands, internet research and hands on clinical laboratory practice.

Examination Standards

Students are expected to be honest in all academic work. Cheating of any kind is not tolerated. Violations of this principle will result in a grade of zero in the course and additional disciplinary action including expulsion as determined by CURE.

COURSE OBJECTIVES:

Upon successful completion of this course, the student should be able to:

- Describe the preparation of the patient for a gynecology sonogram!
- Identify the appropriate transducer for a gynecology ultrasound.
- Describe the embryology, anatomy, function, and normal sonographic appearances of the female pelvis.
- Discuss the sonographic techniques, including emerging technologies, used to evaluate the female pelvis.
- Identify the clinical indications and laboratory values associated with abnormalities and diseases of the female reproductive system.
- Practice the scanning protocols relating to the normal and abnormal female pelvis, normal and abnormal as indicated by ultrasound.
- Discuss the differential diagnosis of all of the following areas: normal and abnormal female pelvis, normal and abnormal 1st, 2nd and 3rd trimester pregnancies and fetal anomalies as indicated by ultrasound.
- Practice the scanning protocols relating to the normal and abnormal female pelvis, normal and abnormal 1st, 2nd and 3rd trimester pregnancies and fetal anomalies as indicated by ultrasound.
- Identify and assess normal and abnormal laboratory values associated with the female pelvis and obstetrics patients.
- Define and use appropriate measures in the evaluation of pregnancy.
- Identify and describe the placenta.
- Define and describe the maternal blood flow and structure of the umbilical cord structure.

- Explain ultrasound safety and the basic premise of as low as reasonably possible (ALARA)
- Discuss the safety of 2D, 3D, and Doppler imaging.
- Successfully perform sonographic evaluation of the female reproductive system and pelvic cavity (uterus, ovaries, and pregnancy) transabdominal and transvaginal protocols independently.
- Pass the scanning laboratory skills related to gynecology ultrasound.
- Recognize and discuss ultrasound results based on class discussions, video, lecture, and PowerPoints.
- Pass the didactic course with a minimum passing grade of 73.

Attendance Policy

Routine, regular class attendance is a requirement for this course.
Promptness and dependability are the soft skills that garner special attention

Attendance Rubric:

Up to 25% will be deducted for each hour missed or for any part thereof (5% for each 10 minutes missed).
Students are required to meet their professor to discuss matters of tardiness or absence.

Students are required to notify their instructor via E mail or voicemail to communicate an absence or lateness from class

A student absent from all of his or her classes for a period of 14 consecutive calendar days during which classes or examinations are in session will be regarded as having withdrawn from the program and will receive a grade of W for each course in which he or she is enrolled.

HOMEWORK

An essential part of every program of study, homework enables students to master course learning objectives and leads toward the achievement of the overall program objectives. Homework refers to self-directed learning activities such as reading assignments, research activities and projects, quiz/exam preparation, knowledge checks and self-assessments, etc.

Late Homework/Make-Up Policy

Late homework will be accepted under the following guidelines:

1. It must be turned in within two business days (Monday through Thursday) after the due date.
2. However, homework turned in after the due date will be penalized 10% for each day it is late up to two calendar days after the due date.
3. Homework submitted after three business days after the due date will earn a zero.

Exams and quizzes - Make up of a quiz or an exam will be allowed under the following extenuating circumstances only:

- Documented illness
- Military duty
- Death of a family member
- Court appearance or jury duty

Documentation of the above circumstances must be provided to the Program Director. ***The ability to make-up a quiz or exam must be approved by the Program Director.***

All assignments should be submitted directly to the instructor.

Classroom Policy

Our goal is to build a respectful learning and work environment that allows for positive communication and teamwork. To support your development, all students must abide by academic policies related to attendance, behavior and professional conduct as published in the current institutional catalog. Special attention is granted to the following items:

- There is to be no eating or drinking in any lab/computer class.
- There is to be no eating in ANY class.
- Cellular phones may only be used during class if/when advised by the instructor and required for a learning activity.
- All students must be in full uniform which includes white or black shoes at **ALL TIMES.**
- **There are no rescheduled exams *unless specified by the professor* with advance notification.**
- The determination to accept late homework assignments is made only by the instructor and occurs only in the event of extraordinary circumstances. If the professor does allow for an extension on assignments, there will be an automatic daily reduction by 10 points, each day beyond the original due date.
- All classes are expected to start and end at a predetermined start and end time.

Accommodations for Students with Disabilities

If you have a special learning need, please contact the Program Director. The Program Director and the Campus President will work with the student to identify reasonable accommodations or adjustments necessary to enable him or her to fully participate in the educational processes.

Professional Responsibility

Students are expected to aspire to the same high level of ethics and professionalism that is demanded of registered Ultrasound students. This requires students to conduct themselves as professionals and to fully engage in the learning process by preparing for and participating in all classroom, laboratory and clinical activities.

Attendance is taken each hour of class by the instructor and turned over to the Registrar before the end of the class day. Early departures, tardiest, and class cuts will be recorded in quarter-hour increments and a proportional grade reduction will also be applied. A period of less than 15 minutes will be counted as a quarter-hour of absence. Attendance records are maintained by the Registrar as part of the student's permanent academic record.

The course syllabus is a general plan for the course; deviations may be necessary and will be announced.

OBSTETRICS SYLLABUS

Below is the course schedule which identifies lecture topics, assignments and other course activities and the homework required for completion each week.

Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class
Week One Objectives: Introduction to course, class policies, and Syllabus • Being able to list indications for obstetric ultrasound exams. • Describe the biologic effects of diagnostic medical ultrasound energy. • Define the ethics in obstetrics. • Define the sonographic characteristics of the yolk sac, embryo, amnion and chorion, and gestational sac. • Describe sonographic measurements being performed in all trimesters. • Describe complications that can occur in the first trimester.		
DAY 1. Role of Sonography in Obstetrics Clinical Ethics for Obstetrics Sonography Normal First Trimester	Power Point Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 47 Role of Sonography in Obstetrics Pages 1161-1170 Chapter 48 Clinical Ethics for Obstetric Sonography pages 1171-1176 Chapter 49 The Normal first Trimester Pages. 1177-1193 Chapter 47 Homework: Workbook Pages 361-365 Chapter 48 Homework: Workbook Pages 367-368 Chapter 49 Homework: Workbook Pages 369-376
DAY 2: First Trimester Complications Sonography of the 2nd and 3rd trimester Obstetric Measurements by Sonography	Power Point Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 50 First-Trimester Complications Pages: 1194-1215 Chapter 51 Sonography of the Second and Third Trimester Pages. 1216-1252 Chapter 52 Obstetric Measurements by Sonography Pages. 1253-1267 Chapter 50Homework: Workbook Pages 377-381 Chapter 51Homework: Workbook Pages 383-392 Chapter 52 Homework: Workbook Pages 393-397
DAY 3. Fetal Growth Assessment by Sonography Sonography and High-Risk Pregnancy Prenatal Diagnosis of Congenital Anomalies	Power Point Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 53 Fetal Growth Assessment by Sonography Pages. 1268- 1277 Chapter 54 Sonography and High-Risk Pregnancy Pages. 1278- 1297 Chapter 55 Prenatal Diagnosis of Congenital Anomalies Pages. 1298- 1313 Chapter 53 Homework: Workbook Pages 399-402 Chapter 54 Homework: Workbook Pages 403-410 Chapter 55 Homework: Workbook Pages 411-416
Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class

Week Two Objectives:

- Describe the embryogenesis and it functions.
- List and describe the sonographic findings of the placenta and the clinical significance of pathology.
- Describe umbilical cord insertion into the fetus and the placenta and the anatomy seen.
- Identify mass or any abnormal with an umbilical cord.
- Lean how to determine the amniotic fluid volume.
- Be able to evaluate for cleft lip and plate.
- Describe abnormalities of the neck.
- Discuss the anomalies that can occur in the fetal head, thorax, spine and limbs.

DAY 4. Placenta Umbilical Cord Amniotic Fluid and Fetal Membranes	Power Point Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 56 Placenta Pages. 1314- 1332 Chapter 57 Umbilical Cord Pages. 1333- 1345 Chapter 58 Amniotic Fluid and Fetal Membranes Pages. 1346- 1363 Chapter 56 Homework: Workbook pages 417-423 Chapter 57 Homework: Workbook pages 425-429 Chapter 58 Homework: Workbook pages 431-436
DAY 5: Fetal Face and Neck Fetal Neural Axis	Power Point Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 59 Fetal Face and Neck Pages. 1364- 1384 Chapter 60 Fetal Neural Axis Pages. 1385- 1405 Chapter 59 Homework: Workbook Pages 437-446 Chapter 60 Homework: Workbook Pages 447-452
DAY 6. Fetal Thorax Fetal Skeleton	Power Point Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 61 Fetal Thorax Pages. 1406-1418 Chapter 65 Fetal Skeleton Pages 1479- 1493 Chapter 61 Homework: Workbook Pages 453-458 Chapter 65 Homework: Workbook Pages 479-485

Week Three Objectives: • Describe the embryology and functions. • Differentiate between different pathology. • Being able to identify what we are seeing on ultrasound.		
DAY 7. Fetal Anterior Abdominal Wall Fetal Abdomen	Power Point Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 62- Fetal Anterior Abdominal Wall Pages. 1419-1432 Chapter 63- Fetal Abdomen Pages. 1433-1447 Chapter 62 Homework: Workbook Pages 459-462 Chapter 63 Homework: Workbook Pages 463-468
DAY 8: Fetal Urogenital System Review	Power Point Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 64 Fetal Urogenital System Pages. 1449-1478 Chapter 64 Homework: Workbook Pages 469-478
DAY 9. SCANOUT/ FINAL	Power Point Presentation/ Lecture/LAB	FINAL EXAM